Some Fundamental Experiments with Wave Guides

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Classic Paper

This paper describes in considerable detail the early apparatus and methods used to verify some of the fundamental properties of wave guides. Cylinders of water about ten inches in diameter and four feet long were used as the experimental guides. At one end of these guides were launched waves having frequencies of roughly 150 megacycles. The lengths of the standing waves so produced gave the velocity of propagation. Other experiments utilizing a probe made up of short pickup wires attached to a crystal detector and meter enabled the configuration of the lines of force in the wave front to be determined. This was done for each of four types of waves. For certain types the properties had already been predicted mathematically. For others the properties were determined experimentally in advance of analysis. In both cases analysis and experiment proved to be in good agreement.

Previous papers [6], [7] have set forth the mathematical theory and a few of the experimental results concerning the transmission of electromagnetic waves through hollow metal tubes, through pipes filled with insulation, and also through cylinders of dielectric material. The purpose of this paper is to describe in considerable detail the methods by which the verifying experiments were made and to add data which it was not feasible to present previously. The methods described are believed to be of interest not only for the way in which they are able to verify the theory but also because they seem to be pointing toward a new technique of electrical measurements.

The previous papers have described four of the many forms of waves that may be propagated through guides. These waves may be distinguished by such propagating characteristics as velocity and characteristic impedance and also by the various configurations of the lines of force which go to make up their wave fronts. Also there is a critical or limiting frequency below which power is not transmitted. For convenience of reference certain of the standard configurations have been reproduced as Fig. 1 below for the particular case of a hollow metal pipe of circular cross section. These four configurations have been designated rather arbitrarily as E_0, E_1, H_0, and H_1 waves, respectively. Somewhat similar waves are also possible in wires of dielectric material where no metal shield is present but in that case the lines of force which are shown above as terminating on the conducting boundary extend into the surrounding space and close as loops.

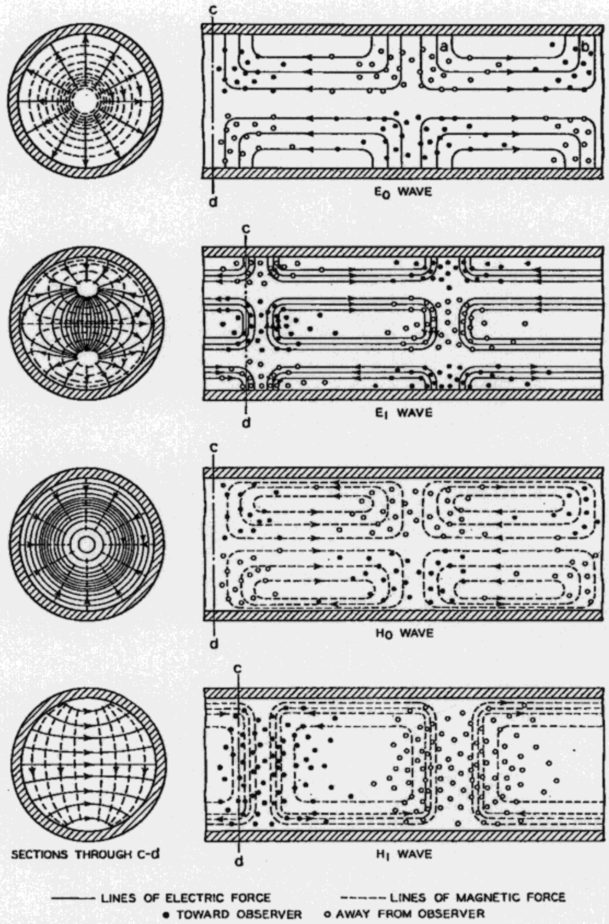

Fig. 1. Approximate configuration of lines of electric and magnetic force in a typical wave guide. Small solid circles represent lines of force directed toward the observer. Propagation is assumed to be directed to the right and away from the observer.

Obvious questions that might be asked at this time are: 1) Do the nice geometrical configurations shown in Fig. 1 actually exist in practice? 2) How are these waves launched

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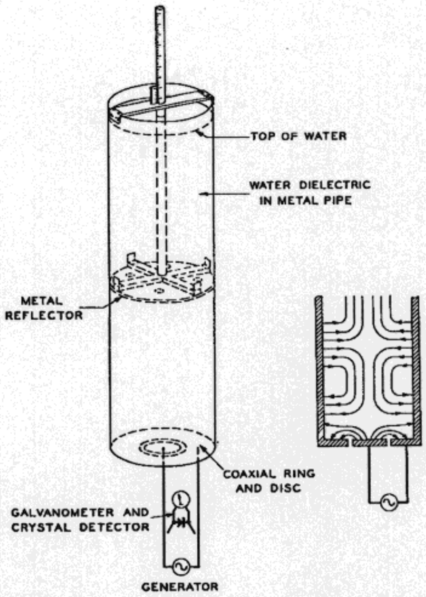

Fig. 2. Schematic of apparatus used to test theory of propagation through wave guides.

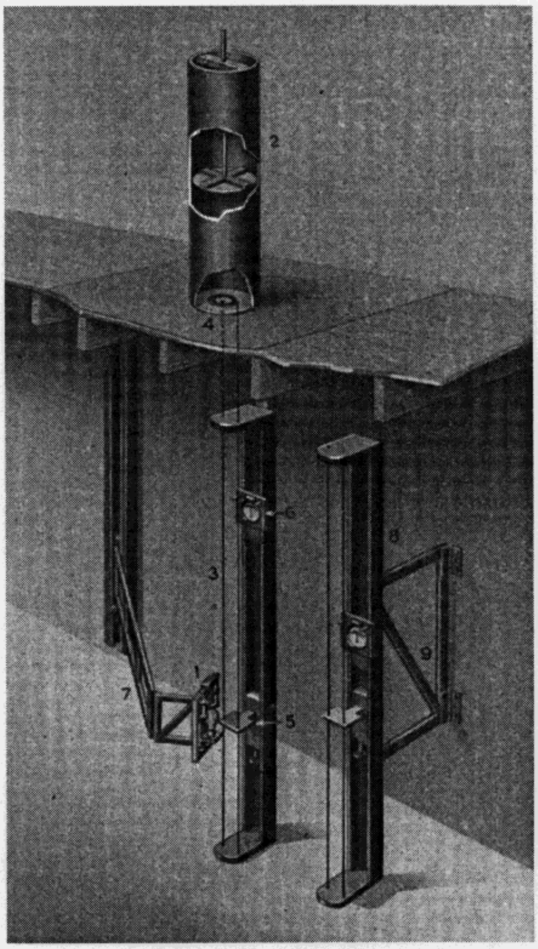

Fig. 3. Arrangement of oscillator, Lecher wires, and water column used in testing the theory of waves in dielectric guides.

and received? 3) How can their velocities be measured? The following paragraphs are directed at these answers.

I. Experimental Verification

The verification of the above-mentioned properties of waves and their guides has involved two rather different ranges of frequencies. In the earlier work which is the particular object of this paper, frequencies between 100 and 400 megacycles were used. In this case the waves were propagated through cylinders of water which served as the experimental guides. More recently it has been feasible to use air-core guides. However, this has called for much higher frequencies. The latter are of the order of 1000 megacycles (λ = 30 centimeters) and 4000 megacycles (λ = 7.5 centimeters). The results obtained by the two methods are, however, very similar so that the more recent work can be said to be largely confirmatory.

II. Apparatus

The apparatus used in this preliminary study is shown in very simple schematic form in Fig. 2 and in greater detail in Fig. 3. As already explained the dielectric guide under observation consisted of a cylinder of water about four feet long. In the course of the experiments four such columns were used. Two were enclosed in copper tubes ten inches and six inches in diameter, respectively, and two were within bakelite tubes of these same diameters. The bakelite was considered sufficiently thin and its dielectric constant so low compared with water that the whole could be regarded as a cylinder of water only.

Waves generated by the source (1) (Fig. 3) were set up in the water column (2) through the intermediary of tuned parallel wires (3) to which the water column was only loosely coupled. The kind and the degree of coupling could be varied through the use of different arrangements at the base of the column. These were sometimes referred to as "pads." In some cases this coupling was analogous to a mutual capacitance and in others it might be regarded as conductive or inductive. The shape of the conductors on the pad determined the type of wave produced. When the E_0 type of wave was desired the pad took the form of metal disk surrounded by a metal ring. The two Lecher wires were then connected to the center and outside rings of the pad, respectively. If other types were wanted pads were used made up of other electrode arrangements as illustrated in Fig. 4.

The essential components of the assembled apparatus are shown in detail in Fig. 3. A shielded platform about six feet square separated in a general way the water column from the rest of the apparatus. The removable pads which afforded the coupling made close contact with the shield, thereby minimizing the wave power that might reach the dielectric from spurious sources. This isolation was acceptable but by no means perfect as the whole room was sometimes found to contain standing waves.

The parallel wires were capable of adjustment for length by means of a 4-×4-inch brass bridge (5) which was movable along the wooden supporting frame. A permanently mounted scale enabled the position of the bridge to be read. A 0- to 200-microampere meter (6), to which was

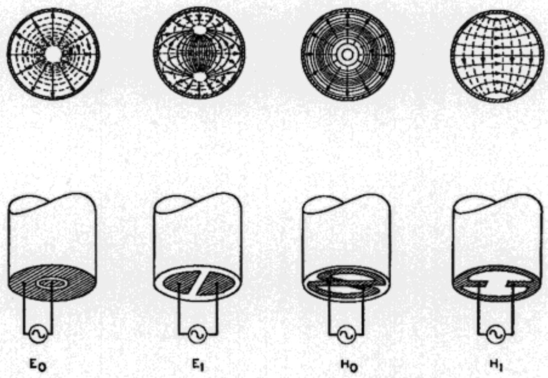

Fig. 4. Conductor arrangements used in launching various forms of guided waves.

attached a sensitive crystal detector, was loosely coupled to the parallel wire system. This showed conditions of resonance in the parallel wires. This parallel wire system proved to be a moderately accurate wavemeter but it was not relied upon for this purpose. Its main function was that of impressing on the end of the dielectric guide waves of a definite configuration. This was considered to be more feasible than to couple the oscillator direct to the column.

The source (1) was mounted on the end of a wooden arm (7) hinged at two points. This permitted a wide range of couplings to the parallel wires. In some instances the separation was but a few inches but in other cases it was two or more feet. This arm was also capable of vertical displacement in order to meet the wide range of conditions due to wave length changes. The power leads to the oscillator were cabled and laced to the arm. There was little trouble from standing waves on these leads.

An independent wavemeter system (8) also made up of parallel wires was arranged vertically on a second hinged arm (9). This would be brought in from the right when it was desired to measure wave length. In this case the wires were much larger and the mechanical arrangement was such that more accurate bridge settings could be made than for the case of the parallel wires described above. This arrangement permitted measurements when the oscillator was under normal load. Its sensitivity was such that it led to no appreciable reaction on the source.

Standing waves were detected in the water column by three rather different methods. Substantially the same results were obtained by all. In one case the parallel wire system was adjusted for resonance, as shown by the meter (6), with no water in the column. Water was then admitted slowly through a hole in the bottom of the column. As the level of the water rose the deflection of the meter varied periodically through rather wide limits, indicating points of resonance in the column with a corresponding absorption of power from the parallel wire system. When the coupling between the column and the parallel wire circuit could be made small, the reaction between the two was also small except at points close to where absorption took place. This method is very similar, of course, to the scheme that was at one time generally used in radio measurements where resonance in a secondary circuit was indicated by

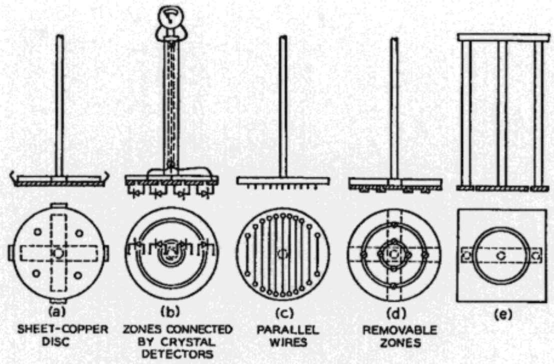

Fig. 5. Various arrangements used to reflect guided waves.

absorption in a primary. In this experiment the distance between successive absorption points was taken as one half of the wave length as measured in the medium.

In another method of detecting standing waves, the column was nearly filled with water and the parallel wires again adjusted for resonance. A circular disk of sheet copper which contacted the sides of the column was lowered into the water. This showed the same reactionary effects on the primary circuit as that of the changing water level. In the third method which was used in connection with E_0 waves, the copper reflector was divided into three coaxial rings between which "waterproofed" crystal detectors were connected. This is shown in Fig. 5(b) below. Connecting wires from the detectors were brought out to an external microammeter which indicated maximum conditions. The second of these three methods was most generally used.

Fig. 4 shows various coupling pads that were used. Each of the four pads was designed respectively for one of the types of waves shown in Fig. 1.

Fig. 5 shows some of the reflectors. That shown in (a) was most generally used. It would of course reflect any of the principal types of waves. That shown in (c) was used to investigate the state of polarization in H_1 waves. That shown in (d) had removable sections and indicated what zones of the cross section of the guide contained the greatest E_0 wave power. That shown in (e) was used to determine the magnitude of fields outside of the guide itself. Two other reflectors not shown made up respectively of radial wires and wires arranged as coaxial circles were used in investigating the orientations in E_0 and H_0 waves.

A meter and detector combination similar in principle to that used on the wavemeter served as a probe in determining the direction and intensity of the field in and about the dielectric wire. Fig. 6 shows the details of both the probe and its bakelite mounting. The latter was clamped to the top of the water column and adjusted so that the probe wires just reached the water. The probe could rotate about its axis as well as be moved laterally along the slot shown. The part in which the slot is located was also free to rotate with respect to the column so that the probe could be carried over the entire surface of the water. The two rotating parts were laid off in degrees and the slot carried a centimeter scale. Typical data obtained by this method are shown in Figs. 11, 12, and 13 below.

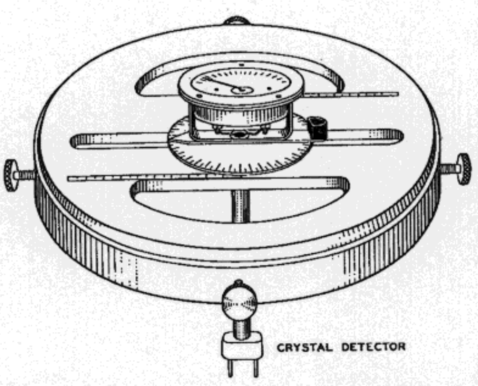

Fig. 6. Crystal detector and meter used as a probe in verifying the arrangement of the lines of electric force in a wave guide.

III. Experimental Results

The primary purpose of these experiments was to establish the existence of guided electric waves, particularly for the case of a guide surrounded by a conductor. This simple objective was readily accomplished as standing waves of a kind were produced with little or no difficulty. However, the waves first found turned out to be rather complicated usually indicating a mixture of two or more types each traveling with different velocities. It then became necessary to disentangle the component parts.

It is not feasible to discuss here all of the difficulties that were encountered except to say that, unless certain precautions are used in the design of the launching mechanism, many spurious waves may be set up in a guide. In most cases, however, these spurious waves may now be identified as a mixture of two or more of the standard types. The various coupling pads shown in Fig. 4 were evolved from some of these early experiments with a view to a fairly pure wave of the desired kind. It was in connection with this problem that the simple probe meter shown as Fig. 6 was first used. This disentangling process led to forms of waves not at first appreciated. However, mathematical analysis was soon able to account for these and other waves as well.

The method for determining velocity of propagation using the principle of standing waves was not essentially different than that commonly involved for the velocity of sound waves in air columns or for electric waves on Lecher wires or in free space. It is based on the simple fact that two oppositely directed waves of the same length and approximately the same amplitude give rise to a series of successive maxima and minima that appear to be at rest. Measurements of the distances between alternate maxima (or minima) give wave length λ_g as observed in the guide. This together with a knowledge of the frequency f permits velocity in the guide to be calculated by the simple relation $V_g = f\lambda_g$.

After methods had been found for producing a relatively pure wave, the measurements of the lengths of standing waves in the guide were generally consistent. Such measurements were subsequently made under a large range of conditions for each of the four water columns described above and for each of the types of waves under study. Typical amongst the results is the curve for the E_0 wave

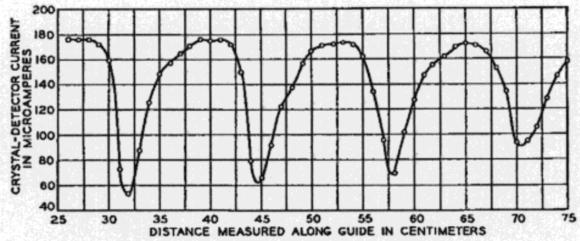

Fig. 7. Typical standing waves in a dielectric guide.

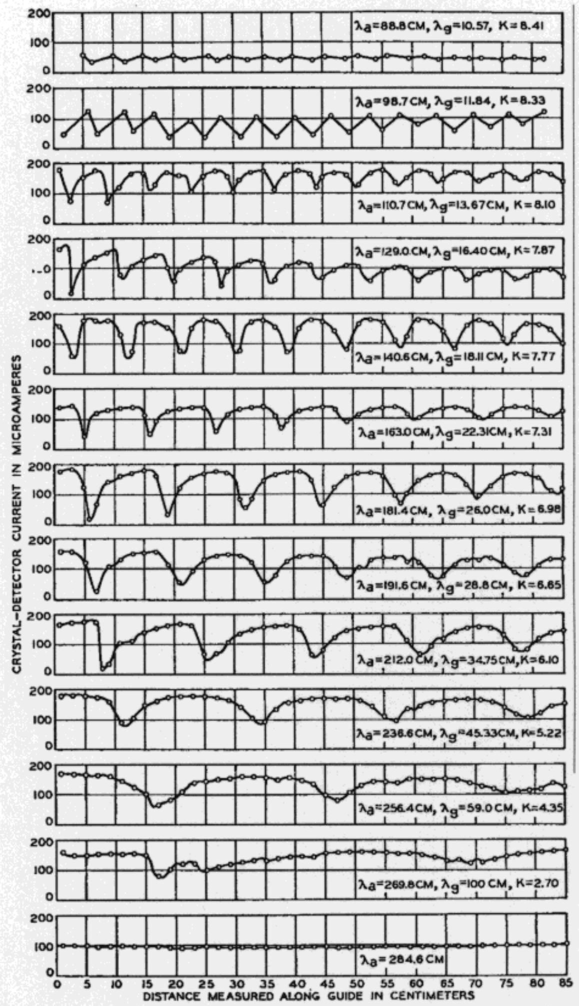

Fig. 8. Experimentally determined standing waves in a ten-inch diameter column of water surrounded by a copper sheath.

shown as Fig. 7. This was plotted from corresponding readings of the reflector levels as measured above the bottom of the column and deflections of the indicating meter coupled to the parallel wires. In order to show how these effects vary with frequency, several of these curves are shown in Fig. 8 for the case of ten-inch diameter copper column.

In general, two limitations were encountered in this work. At the shorter wave lengths the minima were so close together that only the extremes could be located definitely. At the longer waves insufficient minima were included in the relative short columns at hand to measure wave length. This latter limitation, unfortunately, prevented observations

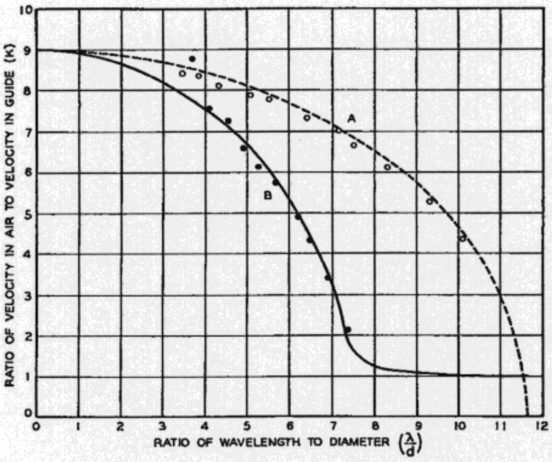

Fig. 9. Relative wave slowness in water columns. (A) Water enclosed in a copper cylinder. (B) Water enclosed in a bakelite cylinder.

near the critical or cutoff wave length. This difficulty was partially overcome by a different type of observation which will be described later. The close coupling which was necessary to give a readable deflection at the shortest wave lengths indicated high attenuation and strongly suggested that the operating frequency was then near an absorption band for liquid water.

It will be noted that corresponding to each curve in Fig. 8, three values are recorded. The first is the wave length λ_a, measured in air. The second is λ_g, or the wave length measured in the guide, while the third is a ratio of these quantities and is designated as K. The latter may be regarded as the relative slowness at which waves travel in the medium as compared with free space and is a quantity susceptible to calculation from the theory of guided waves. The orderly manner by which the relative slowness approaches the value nine, which is the value assumed for the index of refraction of water, is made more evident by the upper curve of Fig. 9. The points are experimental. The curves result from a calculation of wave slowness.

Similar but less extensive measurements of λ_g and K for E_0 waves were made for other columns both with and without metal shields with results somewhat similar to those shown in Fig. 8. These latter curves are not regarded as being of sufficient general interest to be reproduced here. The resulting values of K are, however, plotted in Fig. 9. It will be noted that the experimental data agree rather closely with those calculated. A comparison of the two curves corresponding respectively to a dielectric guide with and without a sheath indicates that in general the velocity of propagation is greater without the sheath than with the sheath. This may be interpreted to mean that when there is no sheath the lines of electric force extend into the surrounding medium which in this case is air so that the resulting velocity is dictated not only by the dielectric constant and the dimensions of the guide but also by the properties of the external medium as well. The underlying theory also points to this view; Fig. 10 is a composite assemblage of the available data relative to a shielded guide

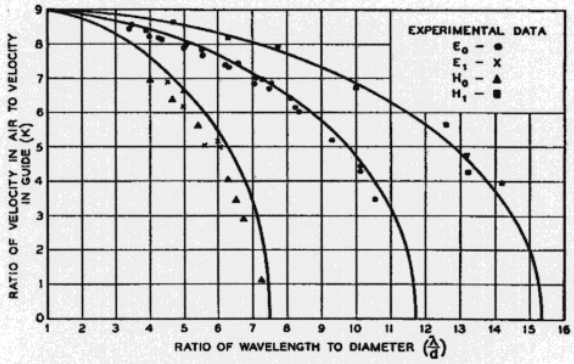

Fig. 10. Relative wave slowness in a copper sheathed water column for each of four principal types of waves.

using water as the dielectric material. It needs no further comment.

IV. Limiting Cases

The data shown in Fig. 8 together with the values of K plotted in Fig. 9 indicate quite clearly that two limiting cases are being approached. The first is for $K = 9$ where the wave length in air is relatively short and the second is for $K = 0$, where this wave length approaches a critical value determined by the diameter of the guide. This corresponds of course to the critical frequency below which no substantial amount of wave power may theoretically be transmitted through a dielectric guide. Also it is at this frequency that the velocity of propagation approaches infinity. Under these circumstances the wave length becomes somewhat greater than the length of the test columns at hand. It was this limitation that made the method described above inadequate to determine the characteristics near the critical wave length.

Another approach to the behavior of guides near cutoff was made by a somewhat different method. To this end the coupling between the oscillator and the Lecher wires was kept constant. The electromotive intensity impressed on the guide therefore remained roughly constant. The probing meter described above was next arranged to give a convenient deflection. A reading of this meter was taken for each of several exciting frequencies as cutoff was approached. This was regarded as a rough measure of the amplitude of the received wave. It was, of course, necessary to adjust the parallel wires to resonance each time after the frequency had been changed.

These experiments showed that the cutoff property may be readily demonstrated in a qualitative way. When the core of the guide is air the cutoff frequency is very sharp and is close to that calculated. However when the core is water the cutoff frequency is as might be expected somewhat less well defined.

V. Experiments Confirming the Field Distribution

As already mentioned, a probe consisting of a crystal detector and microammeter such as shown in Fig. 6, may be used to determine the direction of the lines of electric force in the wave front. This proved to be very useful

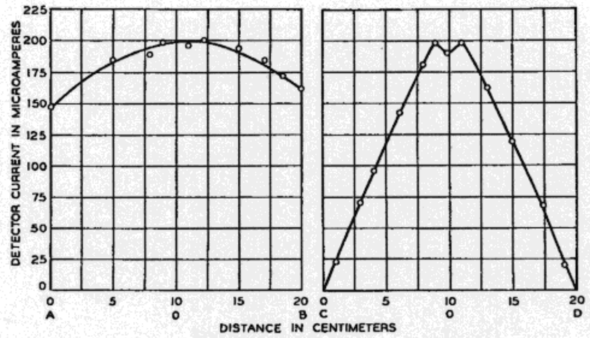

Fig. 11. Measured electric field intensity in the wave front of the H_1 wave. In all cases the probe wires were in the direction of greatest meter deflection.

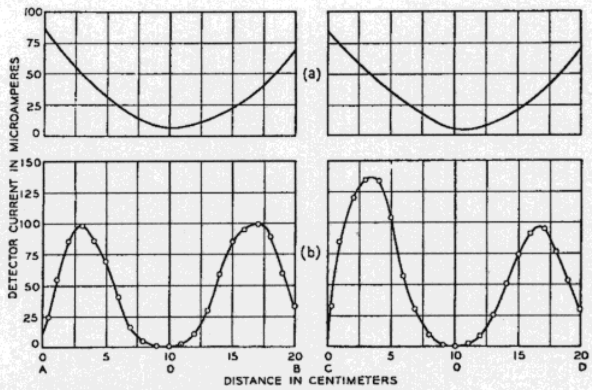

Fig. 12. Measured electric field intensity in the front of the E_0 wave. (a) Water column surrounded by copper sheath. (b) Bakelite sheathed column of water.

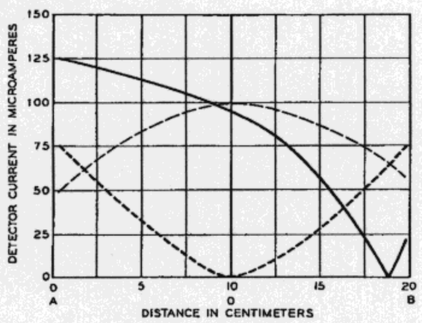

Fig. 13. Typical field distribution when spurious waves are present. Curve (3) represents distribution observed. Curves (1) and (2) represent components going to make up the distribution observed.

not only in verifying the standard forms of waves but also in studying the nature of complex or spurious waves. Typical results are shown in Figs. 11, 12, and 13 below. In these diagrams AOB and COD refer to two mutually perpendicular diameters of the guide along which the probe is carried. The diameter AOB is that along which the electromotive intensity is applied.

In exploring the field of a pure H_1 wave it was readily verified that the lines of force were chords generally parallel to the plane along which the electromotive intensity was applied. When the probe was carried across the diameter AOB the deflection was optimum only when the line connecting the probe wires was kept parallel to this plane and was zero or very small when oriented at right angles thereto. When the probe was carried along COD again the deflection was optimum when parallel with this plane. Fig. 11 indicates roughly the magnitudes of the deflection as the probe was carried along these two diameters. At points near the wall there was a definite tendency for the optimum position to be radial. This is of course in keeping with the idea that small areas of conductors may be regarded as equipotentials and that lines of electric force approach such conductors perpendicularly.

For the E_0 type of wave the probemeter reads a maximum when the line connecting the two probe wires is radial. A typical variation in meter reading as one carries the probe along two diameters of a shielded guide is shown by Fig. 12(a). Fig. 12(b) shows similar data for an unshielded dielectric cylinder.

Similar measurements on the H_0 type of wave in a sheathed guide gave results roughly the same as for E_0 waves in an unsheathed guide except that the line of the probe wires was kept in a tangential direction for maximum deflection. In practice it is not always possible to obtain an altogether symmetrical pattern in either of the E_0 and H_0 experiments. This is evidenced by Fig. 12.

No great effort was made to verify the shape of the E_1 wave. However, it was not difficult to establish the two null points that are evident from the configuration shown in Fig. 1.

Fig. 13 shows a typical distribution of field as found in a wave front where there is a mixture of E_0 and H_1 waves. In an early search for the E_0 wave a mixture of this kind persisted for some time. The results are now easily accounted for by combining say Fig. 11 with Fig. 12(a). Certain other spurious distributions were occasionally found in these experiments which might be ascribed to particular conditions of the setup or even to the presence of the observer himself. The latter was particularly true in cases where no metallic sheath surrounded the column.

VI. Other Experiments Confirming the Field Distribution

A second method throwing some light on the field distribution in E_0 waves consisted of a measurement of the amplitudes of standing waves as reflection took place from different zones of the dissectible disk shown as Fig. 5(d) above. The results of one such test made near the lowest critical frequency are shown in Fig. 14. It will be noted that curve (a) produced by a solid reflector has well defined minima as has also curve (b) which was produced by all zones of the dissectible reflector combined. It appears from (c) that little loss is entailed by removing the central zone. This is substantiated by curve (d) which shows that the central zone alone produces little reflection. Curve (e) indicates that the outside zone plays a very important role in intercepting wave power but it is considerably less important than the intermediate zone as will be seen from curves (f) and (g). This method of studying the field

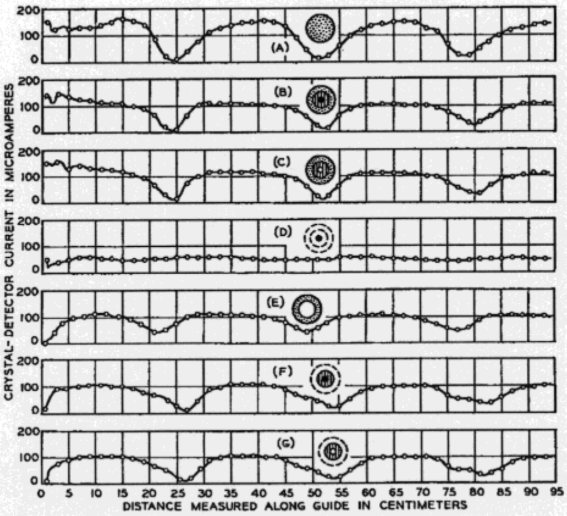

Fig. 14. Relative amplitudes reflected from various zones of a demountable disk.

distribution provided a very interesting check on the probe tests but it is perhaps too cumbersome for general use.

In studying H_1 waves a reflector made up of parallel wires appeared fully as effective as a disk of solid metal so long as the wires were kept parallel to the lines of electric force. When they were rotated ninety degrees there was little or no reflection.

In another experiment a dipole having a length slightly less than the diameter of the water column was constructed by attaching wires to the two terminals of a flashlight bulb. This was mounted on the end of a wooden rod so it could be lowered into the water. At a point near the bottom of the column the lamp would light to full brilliancy so long as the dipole was in the direction of the lines of electric force. As the dipole was rotated still keeping it at this level the brilliance was reduced more or less as the cosine of the angle. Lifting the dipole up and down while oriented to the optimum angle showed the presence of nodes and loops of standing waves. The brilliance was progressively less at each successive loop indicating that the water gave rise to considerable attenuation.

The reflector made up of radial wires was very effective for E_0 waves but it had little or no effect on H_0 waves. In a similar way the reflector made up of circular wires mounted coaxially reflected the H_0 waves but it was indifferent to E_0 waves. This is of course what might be expected and is offered here only as a further confirmation of the predicted theory.

As might be expected, certain of these screen arrangements have proved to be useful devices in subsequent wave guide work. For instance, it sometimes happens that spurious components are present when the H_0 type of wave is generated. If such is the case we may interpose a screen made up of radial conductors and thereby discourage unwanted components. Similarly unwanted components may be filtered from E_0 waves by the use of strainers made up of coaxial metal circles.

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